

The Cycles Agroecosystem Model – Supplemental Information

SI Section I. Model Structure

The Cycles executable resides in a folder selected by the user. The folder must contain an “input” and an “output” folder, with those exact names. The input folder contains all input files. The name of the ControlFile is the simulation ID. When the model run finishes, the output files are printed and stored in a subfolder in the output folder with the ControlFile name (\Output\ControlFileName). Figure SI.1 shows the input and output file structure and dependences. Details of the content of all files can be found at <https://psumodeling.github.io/Cycles/>, alongside a link to the model releases, sample simulations, and scripts to download weather data.

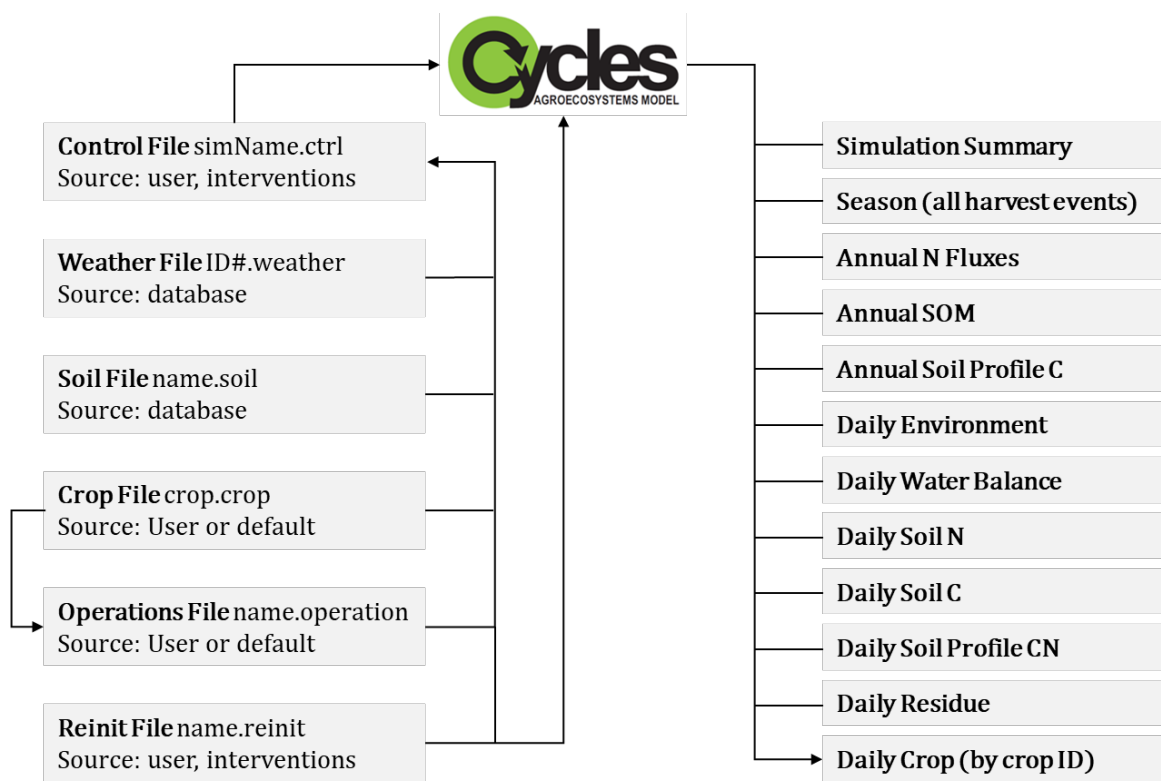


Figure SI.1 Input and output structure of Cycles. The Cycles logo represents the executable. The column to the left shows the relation between input files. The column to the right lists the output files.

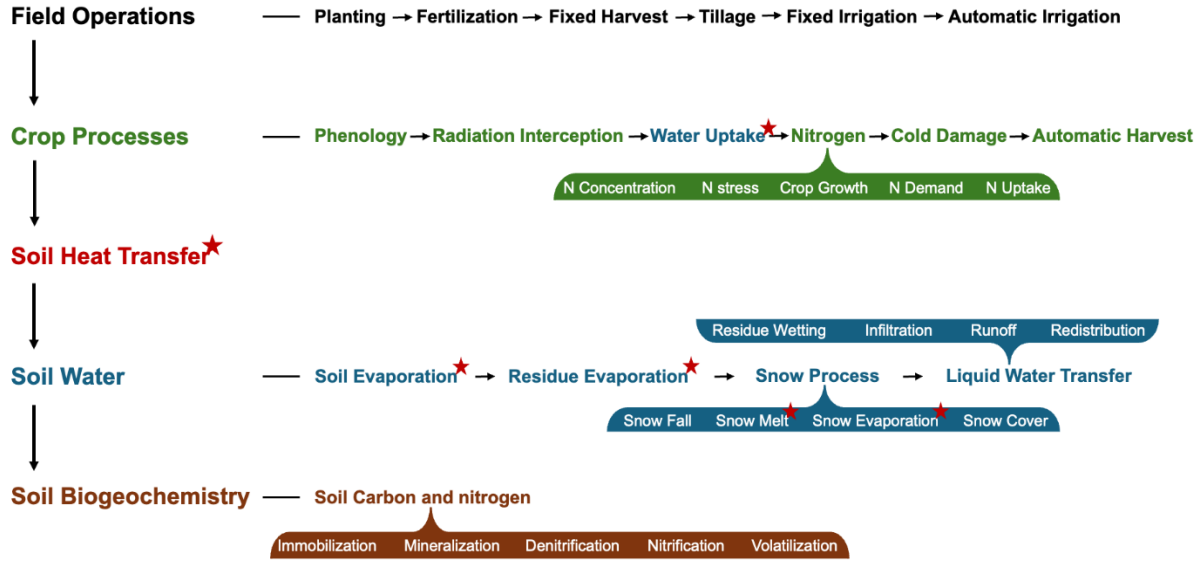


Figure SI.2 General code structure of Cycles simulated processes. The code is organized so that all processes and operations are executed daily (except soil water redistribution). The red star identifies calculations that are part of the energy balance. The first step is to execute all field operations. Tillage mixes the soil and therefore changes many soil state variables. Then if one or more crops are alive, the model simulates all crop processes. This is followed by a simulation of soil processes including soil heat storage, soil water processes (we include the snow layer here), and soil biogeochemistry.

SI Section II. Simulation of vegetation transpiration

For a single species, the maximum evaporation occurring from the green canopy, or potential transpiration (TR_p , $\text{kg m}^{-2} \text{d}^{-1}$), is calculated as:

$$TR_P(\varepsilon_{ie}, k_c, ET_o) = (1 + (k_c - 1)\varepsilon_{ie})\varepsilon_{ie}ET_o \quad \text{Eq. SI.1}$$

The multiplier ε_{ie} is the fractional solar radiation intercepted by the canopy. A temperature factor (from 0 to 1) further adjusts TR_p to account for stomata closure at suboptimal temperatures as photosynthesis decreases. The decrease from a threshold temperature to a base temperature at which transpiration is zero (both input parameters) is linear. At elevated temperature, stomata closure due to high TR_p also limits growth (see below). The calculated TR_p is an input to the algorithm to simulate transpiration (TR , $\text{kg m}^{-2} \text{d}^{-1}$).

SI Section III. Simulation of runoff and infiltration

Infiltration rate (IN , $\text{kg m}^{-2} \text{d}^{-1}$) is the difference between the water reaching the soil surface (W_{in} , $\text{kg m}^{-2} \text{d}^{-1}$) and runoff Q . In the CN method, Q depends on the precipitation volume and soil surface properties. These surface properties are lumped in the “curve number” (CN) for a soil, and a surface water retention term S . The runoff is calculated as follows:

$$Q(W_{in}, S) = \frac{(W_{in} - 0.2S)^2}{W_{in} - 0.8S} \quad \text{for } Q > 0.2S \text{ else } Q = 0 \quad \text{Eq. SI.2}$$

In the original formulation (USDA-SCS, 1972), the surface storage term S depends on soil hydrologic condition and antecedent moisture. These assumptions are lumped in the following empiric equation:

$$S(fs, CN) = 254f_s(100/CN - 1) \quad \text{Eq. SI.3}$$

Thus, if $CN = 100$ (maximum number) all water reaching the soil surface runs off. The constant 254 converts to millimeters. The f_s factor accounts for the surface slope. The base CN for a soil profile is defined by the user or loaded from a database. An algorithm in the model adjusts the CN between dry and wet bounds based on current soil moisture so that a soil profile generates more runoff when wet than when dry. The slope adjustment to S term for runoff calculation reduces the storage capacity when slope is large but has a minor effect for slope below 5%. It is calculated as follows (Williams et al, 2012):

$$f_s = 1.1 - \frac{slp}{slp + e^{(3.7 + 0.02slp)}} \quad \text{Eq. SI.4}$$

Curve number adjustment based on soil moisture factor f_{wc} and an upper and lower CN bound. The factor is 1 for a soil saturated to a depth of 0.6 m and decreases to zero if the soil is air dry. The soil moisture is weighted based on depth, with the soil surface having the most importance. The CN_b is the based CN input in the soil file. The CN used to calculate runoff is therefore:

$$CN_{dry} = \frac{CN_b}{2.3 - 0.013 \times CN_b} \quad \text{Eq. SI.5}$$

$$CN_{wet} = \frac{CN_b}{0.4 - 0.006 \times CN_b} \quad \text{Eq. SI.6}$$

$$CN = CN_{dry} + (CN_{wet} - CN_{dry}) \times f_{wc} \quad \text{Eq. SI.7}$$

Section IV. Crop parameters for growth

Partitioning of new growth to shoot and roots

Partitioning of biomass to shoot (f_{st} , kg kg⁻¹) is estimated allometrically with a modified Michaelis-Menten equation that depends on the proportion of the plant age (thermal time units) compared to that at maturity (TT_f), a minimum allocation of biomass to shoot in juvenile stages (f_{st}^i , kg kg⁻¹), a maximum allocation at maturity (f_{st}^f , kg kg⁻¹), and the TT_f at which the allocation is 0.5 (TT_{f50}):

$$f_{st}(TT_f) = f_{st}^i + \frac{f_{st}^f - f_{st}^i}{1 + (TT_{f50}/TT_f)^4}. \quad \text{Eq. SI.8}$$

Harvest index

Under conditions of no stress, the HI increases asymptotically with the proportion of growth that occurred after flowering (f_{pf}) (or after tuber induction). The asymptote is a notional maximum HI (HI_x), obtained through regression. The model allows for reserves remobilization. If there is no post-flowering growth and plants can remobilize all reserves to grain, then $HI = HI_o$ so that HI_o represents the fraction of the biomass at flowering that could be translocated to grain (or other harvestable component). Stress affects the HI directly by reducing the proportion of growth that occurs after flowering but could also do so by altering the sink size and therefore HI_x . The constant k_{HI} modulates the degree of source-sink co-limitation to grain growth as follows:

$$HI(f_{pf}) = HI_x - (HI_x - HI_o) \times e^{-k_{HI} \times f_{pf}}. \quad \text{Eq. SI.9}$$

The three parameters in this equation are set in the crop input file and f_{pf} is computed during the simulation.

Radiation- and transpiration-use efficiency

Table SI.2 shows a summary of growth parameters for the most common crops and forages. They should be considered close to upper values. The temperature control of transpiration automatically affects growth so that a crop sensitive to cold temperatures may not realize a potentially high radiation-use efficiency. Regions or seasons with a high proportion of diffuse radiation may operate at higher radiation use efficiency (ϵ_R) by about 20% (Stöckle and Kemanian, 2008). Transpiration-use efficiency figures are difficult to find in the literature but for a few species, and we rely on interpretation and indirect measures to estimate their value. For example, on average, sorghum has a lower stomatal conductance and lower ϵ_R than maize. Therefore, it is expected that sorghum will have a higher ϵ_w than maize. The sink control of these two parameters is difficult to model and the user needs to assume that it is embedded in the value of the parameters. Using a layered canopy and sub daily time steps can ameliorate some but not all the limitations of using the dual radiation and transpiration approach to compute growth.

Table SI.2 Sample crop parameters. Most upper figures are from Kremer et al (2009) and Stöckle and Kemanian (2008). Transpiration use efficiency are in some cases estimates that deviate from typical values for the species function type and correspond to VPD = 1 kPa and Patm = 101.3 kPa

Species	Radiation use efficiency (ϵ_R) g biomass / MJ solar radiation	Transpiration use efficiency (ϵ_w) g biomass kg ⁻¹ water at 1 kPa
C₃ Grasses		
Barley, oats, ryegrass, wheat, winter rye	1.7	6.5
Rice	1.6	6.1
Orchardgrass, tall fescue, perennial ryegrass, timothy	1.5	5.7
C₃ Broadleaf		
Potatoes	1.8	6.9
Sunflower	1.6	6.1
C₃ Legumes		
Alfalfa	1.6	6.1
Lentils, soybean, red clover, white clover, hairy vetch	1.1	4.5
C₄ Grasses		
Maize, sugarcane	2.2	8.5
Sorghum (grain)	1.8	9.0
Switchgrass, miscanthus	2.0	8.0

SI Section V. Soil carbon and nitrogen balance

The differential equations for the soil microbial C pool (C_m) and the non-living soil organic C pool (C_s) are:

$$\frac{dC_m}{dt} = \varepsilon_c f_E f_H (f_{ra} k_{ra} C_{ra} + f_{rt} k_{rt} C_{rt} + f_{rz} k_{rz} C_{ra} + f_{rm} k_{rm} C_{rm}) - f_A f_E f_T k_m C_m \quad \text{Eq.SI.10}$$

$$\frac{dC_s}{dt} = \varepsilon_c f_A f_E f_H k_m C_m - f_E f_T f_D k_s C_s \quad \text{Eq. SI.11}$$

The factors f_E and f_T scale turnover rates based on the soil environment ($0 \leq f_E \leq 1$) and tillage stimulation ($f_T \geq 1$) of C_s degradation. The factor f_A stimulates microbial turnover based on the size of the microbial biomass to prevent a buildup, i.e., we assume that there is a maximum microbial biomass carrying capacity. Once C_m exceeds 3% of C_s , then f_A increases exponentially and prevents the build-up of live microbial biomass. The factors f_{ra} , f_{rt} , f_{rz} , and f_{rm} (for aboveground residues, root residues, rhizodeposits, and manure) are multipliers that slightly modify the base microbial efficiency (ε_c). These efficiencies are composites whose upper bound is the maximum microbial C use efficiency; operational values in soils are more ambiguous (see Kemanian and Stöckle, 2010; White et al, 2014). We tentatively assume aboveground and root biomass microbial C use efficiency are the same (between 0.33 and 0.44 kg kg⁻¹ depending on the saturation ratio), that manure is retained with higher efficiency ($f_{rm} \approx 1.4 \times f_{ra}$), and rhizodeposits with less efficiency ($f_{rm} \approx f_{ra}/1.4$). A simplified version of the effect of tillage on turnover rate can be found in Kemanian et al (2011). Organic C saturation affects microbial C use efficiency through the factor f_H and the soil C decomposition rate through the factor f_D .

$$f_H = 1 - \left(C_s/C_{sx}\right)^6 \quad \text{Eq. SI.12}$$

$$f_D = 1 - \frac{1}{\left(1 + \left(\frac{4.5 \times C_s}{C_{sx}}\right)^3\right)}. \quad \text{Eq. SI.13}$$

The saturation ratio is therefore a scaling factor of the microbial C use efficiency and soil C decomposition rate. When C_s approaches saturation, the C use efficiency decreases to zero (more C loss as CO₂ per unit of substrate decomposed by microbes), and the C_s decomposition rate increases due to lesser physical protection of organic matter by organo-saturated mineral surfaces. The response to saturation in Cycles matches measurements well (Kemanian and Stöckle, 2010; Pravia et al, 2019), but is not universal. When turnover rates are reduced by very moist, dry, or cold soils (lower f_E), C_s can accumulate. If temperature (or other factor) is low enough to prevent microbial decomposition of above or belowground residues in a yearly cycle, then residues would accumulate like in Histosols. Increased inputs of fresh residues would increase the C_s steady state, but saturation dynamics sets a hard bound for the maximum asymptote independently of C inputs magnitude. In the short term, the model behavior is linear, but under high C inputs it is highly non-linear.

The C:N ratio of the newly formed microbial biomass (CN_{mbn}) is assumed to vary with the C:N ratio of the decomposing residue and the availability of mineral N. The minimum C:N ratio is assumed to be 8.5 and the maximum 14. The C:N ratio of organic matter in the mineral matrix can be lower, particularly in the subsoil. The C:N ratio of the newly formed microbial biomass is computed in two steps. The C:N of the decomposing residue is assumed to always affect the C:N of the microbial biomass, determining new bounds for the C:N ratio of the microbial biomass, always within the 8.5

and 14 ratios. The mineral N in the soil determines where within that range would the C:N ratio end according to:

$$CN_{mbn} = 8.5 + 5.5 \times \left(1 - \frac{1}{1 + (CN_d/110)^3} \right) \times \left(0.5 - \frac{0.5}{1 + (N_{ppm}/8)^3} \right) \quad \text{Eq. SI.14}$$

where CN_d is the C:N ratio of the decomposing residue, N_{ppm} is the mineral N concentration in the soil in units of weight per unit of weight. For example, if the decomposing residue has a C:N ratio of 55 and the soil a concentration of 8 ppm of mineral N, then the C:N ratio of the microbial biomass formed consuming such substrate will be 8.65. Cycles of microbial consumption and predation would therefore force the C:N ratio of SOM towards 8.5 unless there is a steady addition of fresh inputs with high C:N ratio.

SI Section VI. Measuring maize and shrub willow ET using the eddy covariance method

Details of the data reported here are available in Cangiano (2015) and Montes et al (2021). A summary follows. The experimental site is in Centre County, Pennsylvania, near The Pennsylvania State University campus (40° 51' 36.92" N, 77° 47' 50.37" W, elevation 370 m). Wind is mostly from the west along the main axis of the valley. The willow field was planted in 2012. The stand had six cultivars in long strips perpendicular to the dominant wind direction and planted in two blocks. The cultivars were: *S. viminalis* × *S. miyabeana* 'Fabius' 'Otisco,' and 'Preble,' *S. purpurea* × *S. miyabeana* 'Millbrook,' *S. purpurea* 'Fish Creek,' and *S. miyabeana* 'SX61'. The soil belongs to the Hagerstown Series (fine, mixed, semiactive, mesic Typic Hapludalfs) which have good drainage (hydrologic group B). Organic matter content ranges from 2 to 2.4% in the Ap horizon, although it can be lower in eroded areas. By 2014, as evidenced by the 2015 Google Earth image (Fig. SI.3) the stand was well developed.



Figure SI.3. Willow field in 2015 along with a diagram of the wind direction frequency (footprint) centered on the position of the eddy covariance tower. The fetch in the dominant wind direction was 220 m. More details in Cangiano (2015). The panel to the right is a map of the strips of the willow cultivars from Montes et al (2021).

161

162 Maize was no-till planted in the 15-ha field on May 5, 2014. The field had a soybean crop the prior
 163 year that left 1.8 Mg ha^{-1} of organic C as aboveground residues (grain yield 2.5 Mg ha^{-1}), and a
 164 fescue sod for approximately 30 years prior. The maize plant population was 7.4 seeds m^{-2} in rows
 165 0.76-m apart, and the crop fertilized at planting and on 16 June with 150 kg ha^{-1} of N (40/60 split)
 166 as liquid urea ammonium nitrate. Maize was harvested on 21 November and yielded 11.3 Mg ha^{-1}
 167 (dry) with an aboveground biomass of 21 Mg ha^{-1} of aboveground biomass (9.5 Mg C ha^{-1} , excluding
 168 roots).

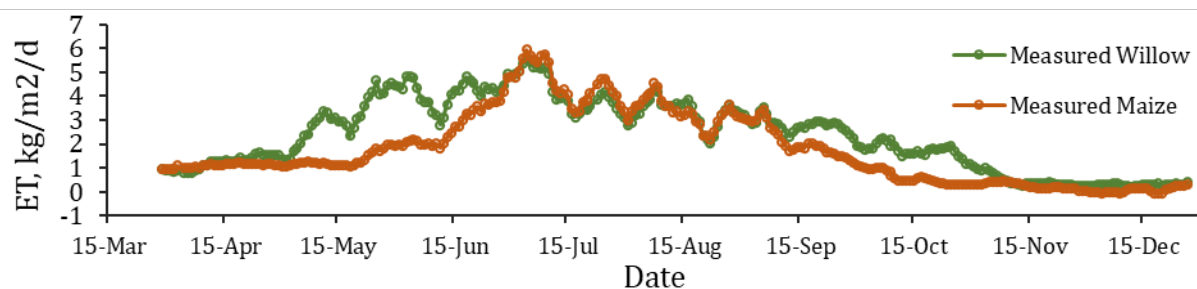
For measuring water vapor and carbon dioxide fluxes using the eddy covariance method, four Campbell Scientific Inc (Logan, UT) systems were installed in pairs of one closed path (CPEC200) and one open path (IRGASON) per field. Deploying both systems simultaneously minimized gap filling. Wind speed was measured with a three-dimensional sonic anemometer with a path length of 10 cm and a frequency response of 40 Hz (model CA27, Campbell Sci.). Air temperature was measured with a 12.7 μ m thermocouple with a frequency response of 40 Hz (model 127 by Campbell Sci.). Water vapor density was measured with an ultraviolet krypton hygrometer with a frequency of response of 100 Hz (model KH20, Campbell Sci.). Net radiation was measured with net radiometers (model NR-LITE2, Campbell Sci.). Soil heat flux was calculated by adding the heat storage in the top 8 cm of soil to the heat flux at 8 cm. Soil heat flux was measured at 8-cm with self-calibrated soil heat flux plates (model HFP01SC, Campbell Sci.) and the heat storage estimated from changes in soil temperature measured with thermocouples (model TCAV, Campbell Sci.). Soil moisture was measured with water content reflectometers (model CS616, Campbell Sci.). The algorithm used to compute the fluxes was the "EdiRe Software for Micrometeorological Applications" provided by Campbell Scientific Inc.

Open path and close path data series from each crop were paired to obtain one series with less gaps for net carbon dioxide ecosystem exchange (NEE), latent heat (LE), sensible heat (H), temperature of the air (T_{air}), net radiation (R_n), mean water vapor concentration (w), and soil heat flux (G). For NEE, LE and H, the pairing was done after the filtering criterion was applied (see below). After pairing, maize NEE and LE were the variables with the highest missing values (54%), while shrub willow R_n has the lowest missing values (18%). Gaps in the EC data series for T_{air} , R_n and rh were filled with NOAA data. NOAA Penn State meteorological station is located near the experimental site (40 43' 12.43", -77 55' 51.06", elevation 376 m). The half-hour comparison of NOAA and EC measurements showed almost perfect correspondence for R_n , vapor pressure and T_{air} (r values of 0.94, 0.92 and 0.99 respectively). For CO_2 , storage flux was calculated based on Aubinet et al.

(2001); this correction affected fluxes minimally. Spikes detection followed Papale et al. (2006). The friction velocity threshold to accept data or resort to gap filling for NEE, H, and LE followed Reichstein et al. (2005). Gap filling for these variables was done with the package REdDyProc (Reichstein and Moffat, 2014). Gap filling for soil heat flux was done estimating the heat flux at the soil surface using an analytical solution for soil heat flux that assumes a sinusoidal temperature variation and needs the daily thermal amplitude, soil heat conductivity and damping depth (between 0.13 to 0.25 m); however, the dumping depth remained relatively constant because the soil was moist most of the year.

There was no closure of the energy balance. The regression (LE + H) vs (Rn - G) yielded a slope of 0.82 and 0.67, intercept of -1.46 and 0.1 MJ m⁻² d⁻¹, and r² of 0.96 (n = 281) and 0.92 (n = 281) for willow and maize, respectively. These figures were used to correct daily fluxes of LE and H.

Figures here show a comparison of measured and modeled ET (Fig. SI.4) and linear relations between NEE, solar radiation, vapor pressure deficit, and ET reference, for the period in which there was full canopy cover in both crops (July and August) (Fig. SI.5). The modeled ET early and late in the season is higher than that measured, but it is possible that measurements underestimate ET when the canopies are leafless or after corn harvest. We cannot resolve that uncertainty, but clearly, the measured and modeled ET track each other when the canopy is fully developed.



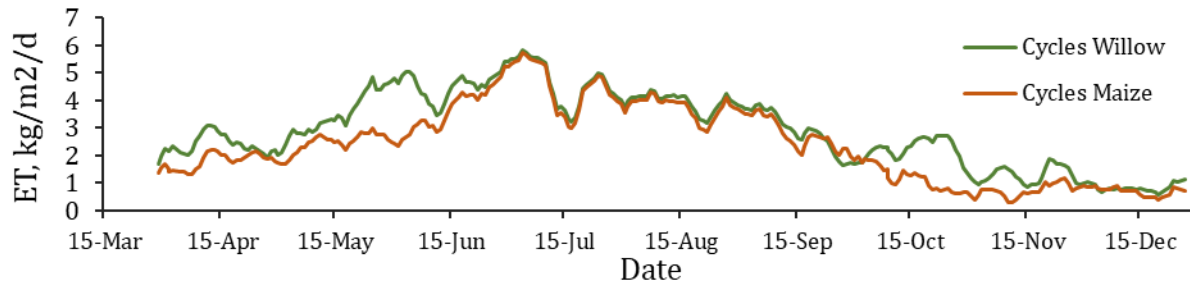


Figure SI.4. Measured and Cycles-modeled ET above adjacent fields of shrub willow and maize at Rockview, Pennsylvania, in 2014. Data presented as centered 7-day moving average. Measured where obtained by the eddy covariance technique and latent heat flux corrected to account for lack of closure in energy balance (Cangiano, 2015).

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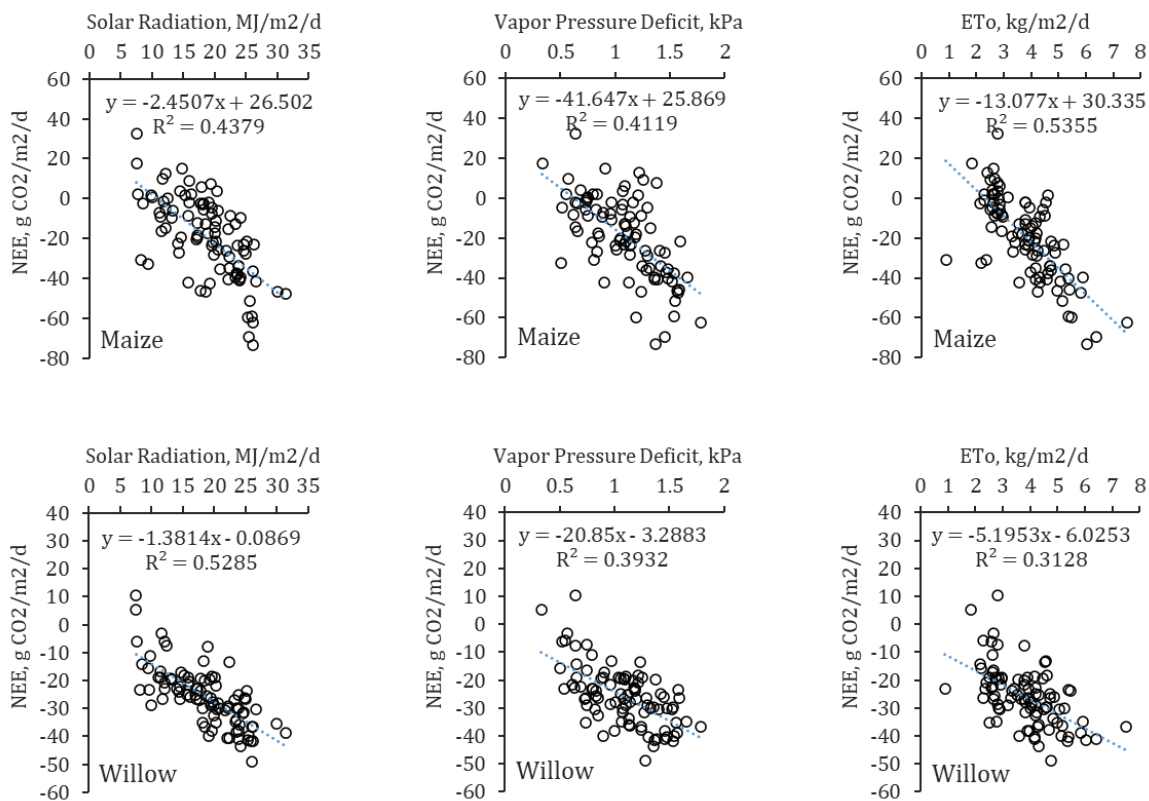


Figure SI.5. Net CO₂ ecosystem exchange for willow and maize during periods of full canopy cover (July and August) plotted against daily solar radiation, daytime vapor pressure deficit, and reference ET.

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213 **SI Section VII. Simulation of spring wheat, spring barley and winter wheat in Eastern Washington**

214 A description of the Cook Agronomy Farm (CAF), part of the USDA LTAR network, can be found in
215 Huggins et al (2014). The 34-ha field (46° 47' N, 117° 5' W; elevation 773-815 m, mean slope 11%
216 and maximum slope 33%) is located 5-km NE of Pullman, WA, USA. The region has a Mediterranean
217 climate, with precipitation averaging 550 mm/year, and is one of the most productive rainfed
218 wheatlands in the USA. Precipitation decreases west towards the lower elevation of the Columbia
219 River and increases East towards the Moscow Mountains in nearby Idaho. The field has a complex
220 terrain, developed over loessian deposits. Soils are of silt loam texture, an association of Palouse
221 (fine-silty, mixed, superactive, mesic Pachic Ultic Haploxerolls)-Thatuna (fine-silty, mixed,
222 superactive, mesic Oxyaquic Argixerolls)-Naff (fine-silty, mixed, superactive, mesic Typic
223 Argixerolls) soil. Agriculturally, the Palouse soils are deep, with root exploration unlimited in the
224 top 2 m, which can store almost 400 mm of PAW. Naff and Thatuna soils have argillic horizons
225 closer to the surface and can, depending on local conditions, limit productivity. Topography can
226 also regulate water distribution and the onset of water stress. In south facing slopes, often with Naff
227 soils that are shallow and eroded, wheat can reach maturity one or two weeks earlier than north
228 facing slopes with Thatuna soils. However, Thatuna soils have a duplex soil behavior, as the argillic
229 horizon can have extremely high bulk density effectively stopping root exploration.

230 Experimental work at the CAF started in 1999, with the planting of the entire field with hard red
231 spring wheat (cultivar WB926R, 14 May 1999), followed by spring barley (cultivar Baronesse, April
232 15). Thereafter, the field was segmented into strips running south-north. The central strip was
233 planted with hard red spring wheat (cultivar Hank, 25 March 2001), and hard red winter wheat
234 (cultivar CDC Falcon, 23 October 2001). Harvest occurred in September in 1999, and in August in
235 all other years. In 1999 and 2000, biomass and yield were determined in a non-aligned grid by

harvesting four adjacent rows at 2-m length (140 samples). In 2001, yield was determined along a north-south transect where spring wheat was fertilized in spring or as split fall/spring with 100 kg/ha of N, or not fertilized (20 2-m² plots). In 2002 similar determinations were on a north-south transect parallel to that of 2001, where the entire field was split fertilized.

In addition to the simulations presented in the main manuscript, Figure SI.6 shows the modeled residual mineral N in 2002. Nitrogen moves down the soil profile after the fall application or is absorbed by plants, but a large fraction of the spring applied N remains in layer 2 at harvest. Low precipitation and high soil water capacitance limit mineral N transport in well drained positions, as simulated with Cycles.

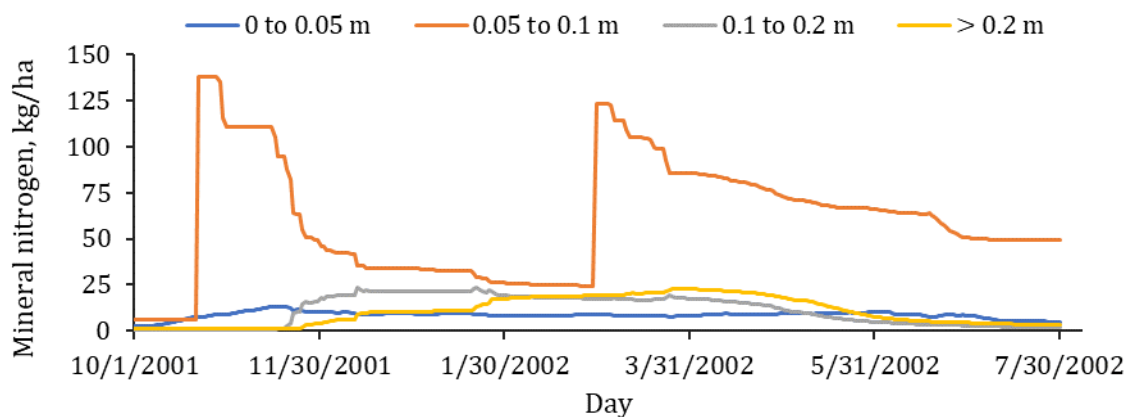


Figure SI.6 Modeled mineral nitrogen from October 2001 to July 2002, bracketing planting to maturity of winter wheat. The two spikes in layer 2 (0.05 to 0.1 m) are fertilization events.

SI Section VIII. Simulations of soybean soil water content

The experiment was conducted at the Penn State Southeast Agricultural Research and Extension Center in Landisville, PA (SEAREC, latitude 40°16' longitude N 77°31' W elevation 110 m). Soybean was planted following winter wheat. The planting was late for the region because the experiment sought to represent extreme double cropping conditions. The experiment had two planting dates

and five replications. The soils are of the Hagerstown silt loam type (fine, mixed, semiactive, mesic Typic Hapludalfs; Duffield silt loam, fine-loamy, mixed, active, mesic Ultic Hapludalfs; and Clarksburg silt loam, fine-loamy, mixed, superactive, mesic Oxyaquic Fragiudalfs).

Weekly soil water content measurements were taken from planting to maturity at SEAREC using a capacitance probe (Diviner 2000, Sentek Pty Ltd, Stepney, Australia) that reported data at 10-cm soil depth intervals. The Diviner access tubes were inserted with a Giddings probe. Since a reliable calibration equation for the Diviner was not found for the research site, the field capacity (θ_{fc}) and permanent wilting (θ_{pwp}) point of each soil layer were calculated with the Saxton and Rawls (2006) pedotransfer function and used to scale the field scaled frequency measurements in between these boundaries as follows:

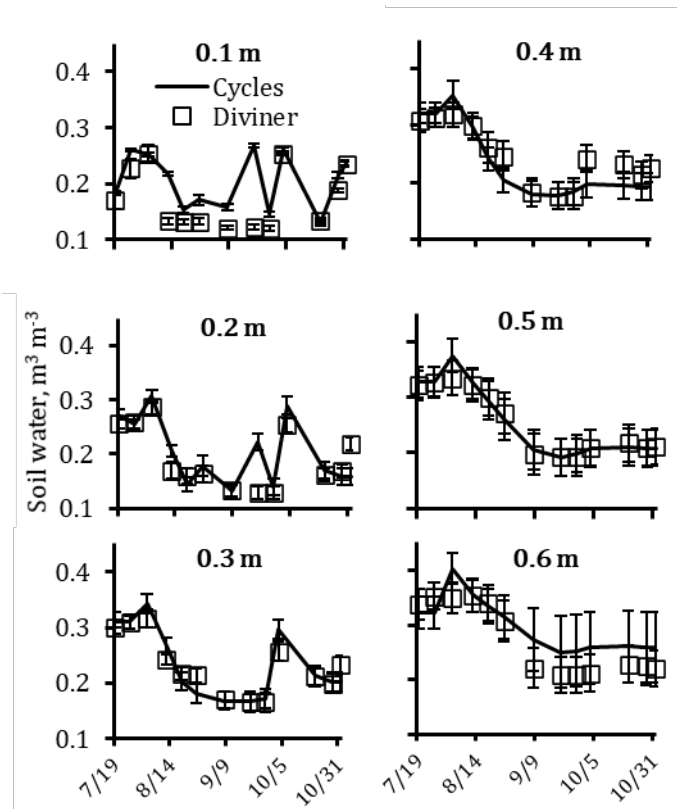


Figure SI.7 Measurement and simulation of soil moisture at the Penn State Southeast Agricultural Research and

Extension Center in Landisville, PA. Soil water content was measured with a Diviner and modeled with Cycles. Data shows soil moisture from soybean a week after emergence to harvest for the second planting date (July 12). Error bars indicate the standard deviations of five replications.

261

$$\theta = \theta_{pwp} + (\theta_{fc} - \theta_{pwp}) \times \left[\frac{R - R_n}{R_x - R_n} \right]^5 \quad \text{Eq. SI.15}$$

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263 where θ is soil water content, R is the raw measured scaled frequency reading, R_n and R_x are the
 264 minimum and maximum scaled frequency reading in the layer, respectively, through the growing
 265 season. This approach was only appropriate because 2016 was a dry year that allowed the soil to
 266 dry until the θ_{pwp} . The power of 5 accounts for the non-linear change in the signal with soil
 267 moisture and was selected to match observed and modeled soil moisture patterns. The goal with
 268 this approach was to compare the measured and simulated soil moisture fluctuations to investigate
 269 if Cycles can capture the seasonal soil moisture variability. Measurements were performed once a
 270 week in one plot in each of the five replications in two soybean planting dates (June 26 and July 12).
 271 Figure SI.7 shows the result for the second planting date in this trial (the main manuscript presents
 272 data for the first planting date).

273

274 **SI Section IX State-wide simulation of every single field in Iowa**

275 Deploying “twin models” of the cropland of the globe is now feasible due to the increasing
 276 computational and data storage capacity. Instead of running practical but simplified versions of
 277 reality using representative soil and climate clusters (Meki et al, 2013), gridded models (Elliot et al,

2015), or emulators (Blanc, 2017), we present the results of simulating >870k fields in Iowa. To set up the simulations, weather data was obtained from NLDAS (North American Land Data Assimilation System) (Xia et al, 2012), soil data from the Gridded National Soil Survey Geographic Database, and field boundaries from a publicly available common land unit (CLU) database, presence of tile drains (Nakagaki et al, 2016), and crop rotations from the CropScape 2013-2016 Data Layers (USDA-NASS, 2022). In addition, a manure distribution algorithm, created using locations and demographics of concentrated animal feeding operations (CAFOs) across Iowa with >1000 animal units (IA DNR, 2016), was used to assign manure to fields within an expanding radius around each facility until the manure supply was depleted.

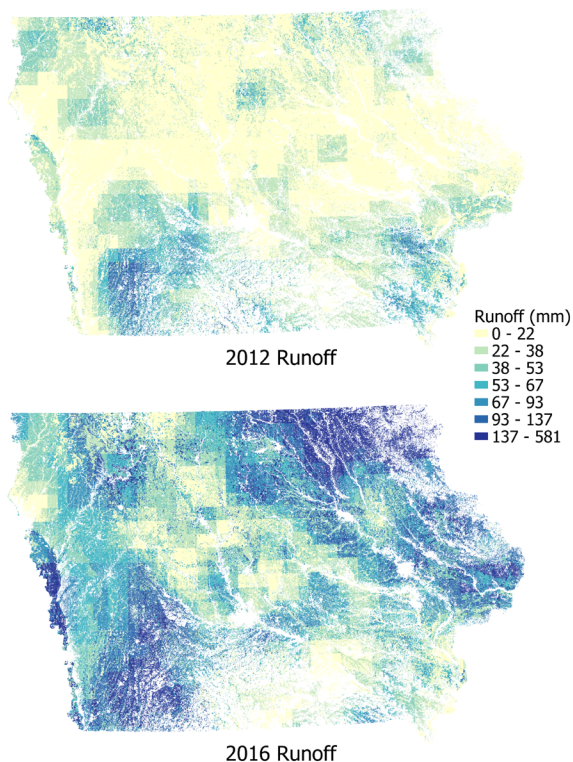


Figure SI.8. Cycles simulated runoff (mm/year) for every field in Iowa in 2012 and 2016.

Only fields with maize or soy were modeled (>85% of the fields in Iowa). Field operations included a fixed planting date for maize or soy, a tandem disk and cultivator pass, no residue removal, and no

cover crop. Nutrient amendments (urea ammonium nitrate or manure N) were applied at a rate of 220 kg/ha of N for maize (N with manure was adjusted upward to account for 0.5 availability compared with mineral N), with a 60 kg/ha of N credit if maize followed soy. If a field was manured then fertilizer was split 50/50 between fall and spring; if UAN (Urea Ammonium Nitrate), it was applied in spring. Simulations were run from 1980 to 2016, with a recurrent crop sequence using the rotation of 2010-2016 for each field. For illustration purposes, we present the simulated runoff in the dry year of 2012 and the wet year of 2016.

The runoff patterns in Iowa differed between dry 2012 and wet 2016, and spatially within each year due to marked differences in precipitation within the state (Fig. 9). The pixelated pattern in Fig. SI8 reflects the NLDAS weather grid; it is expected that the accuracy and resolution of these products will only increase. While the dry year shows an overall low runoff and therefore low water yield, peak hydrographs in small streams can be highly dependent on local conditions, while depth to the water table is more dependent on regional precipitation. The dry areas in 2012 (yellow) were not only affected by drought, but because of the low crop yield, they had residual nitrate that can leach in winter and fall. Tactical approaches to reduce leaching, like initiatives to increase cover cropping with grasses in the event of summer droughts, can be targeted to regions with low summer runoff even if the soil nitrate concentration is not known.

This can be even more critical in areas with CAFOs where manure cannot be easily stored. Using these tools regularly would enable regional and field-tailored analysis with increased ease, fidelity, and utility. The resolution can be increased to include subfields if the quality and compatibility of the inputs allow it, for example with data from high resolution satellite imagery or UAVs (Unmanned Aerial Vehicle) (e.g., Azzari et al, 2019).

SI Section X Cycles-A: Additional outputs obtained with the crop sequence builder

The timeline of the crops selected by Cycles-A in a one-time run for all counties along the 77° meridian within the Chesapeake Bay Watershed is shown in Fig. SI.9. The preponderance of wheat reflects the price selected and the fact that diseases are not included in the model. Notice the increasingly fallow time (white space) northward along the meridian.

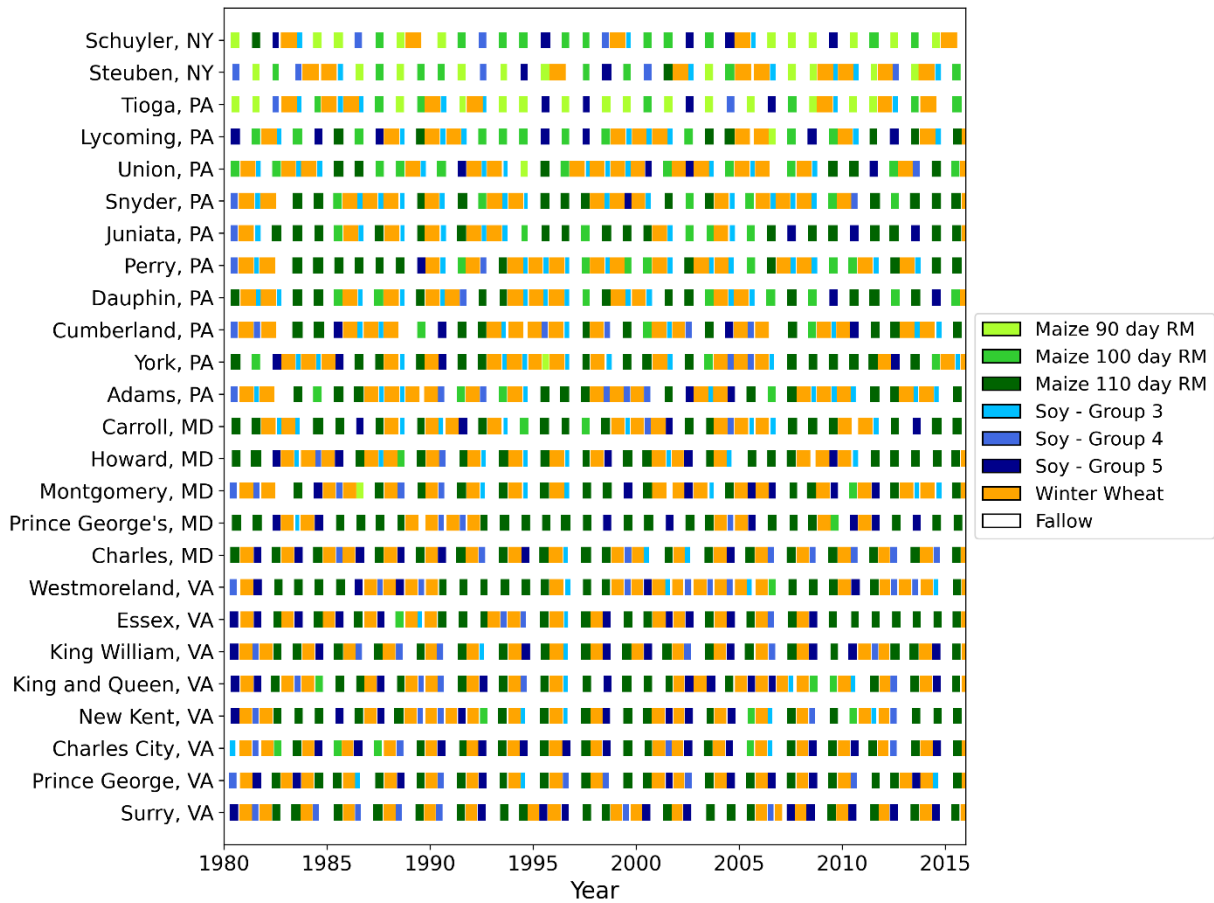


Figure SI.9. Crop sequence along the 77° meridian within the boundaries of the Chesapeake Bay Watershed as selected by Cycles-A, an autonomous crop sequence builder. The horizontal length of each bar reflects the time in the field.

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